2 Goal and Area of Application

This guideline describes general practices of Chemical Biology Consortium Sweden (CBCS) at Karolinska Institutet used to address nuisance compounds in biochemical and cellular assays to avoid misinterpretation of results.

3 Definitions and Abbreviations

PAINS	Pan-Assay Interference Compounds
MOA	Mechanism of Action
(TR)-FRET	(Time-Resolved)-Fluorescence Energy Transfer
SAR	structure-activity relationship
REOS	rapid elimination of swill
NIH	National Institutes of Health
SOP	Standard Operating Procedure
Z´ factor	Statistical measurement of assay quality
QC	Quality Control
IC_{50}	Half Maximal Inhibitory Concentration
EC_{50}	Half Maximal Effective Concentration
HPLC	High-Performance Liquid Chromatography
MS	Mass Spectrometry
HRP	Horseradish Peroxidase
STR	Short Tandem Repeat
TSA	Thermal Shift Assay
DSF	Differential scanning fluorimetry
ITC	Isothermal Titration Calorimetry
Counter-assay	Assay to identify compounds that interfere with the read-out in the primary assay by *e.g.* auto-fluorescence. A counter assay can also be used to identify compounds with undesirable properties such as colloidal aggregation and cytotoxicity.
Orthogonal assay	An orthogonal assay is an assay that uses a different method or principle from the primary assay to measure the same biological activity.
Secondary assay	Follow-up assay used for further validation of the biological effect of a compound by confirming and refining the results obtained from the primary assay

4 Method

4.1 General

A key to avoid nuisance compounds is to consider potential artifacts already during the assay-design and assay-optimization phase. Despite these precautions, a well-defined screening funnel with follow-up assays is crucial to confirm results and eliminate nuisance compounds from the project pipeline. This is an iterative process, allowing for continuous refinement and adjustment.

Nuisance compounds include assay artifacts and molecules with undesired MOAs. Assay artifacts refer to compounds bearing certain properties that can interfere with assay technologies, such as: auto-fluorescence, absorbance, quenching, reporter modulation, chelation, and reagent mimetics. Undesired MOAs that interfere with assay results can be non-target-associated or chemical-based, such as compound aggregation, non-specific reactivity (redox, oxidants, electrophilic functionality), surfactants, metals and impurities; and can also be target/pathway associated or biological, such as cytotoxic, genotoxic, promiscuity/poly-pharmacology, phospholipidosis etc[1-3]. General guidelines to address many of these nuisance mechanisms can be found in the NIH assay guideline[1]. In addition CBCS at Karolinska Institutet have in house guidelines for assay development[4] and quality standards for assay development and screening[5].

4.2 Procedure

Below is a list of examples of assays and in silico evaluations that can be used to identify and remove nuisance compounds. However, this does not exclude the possibility that additional assays may be necessary depending on the specific requirements of the project. The combination of assays and in silico evaluations to be applied in each project should be carefully considered to best address the potential artifacts relevant to the specific context. The selection of an appropriate evaluation pipeline should be determined through a collaborative effort between the responsible biologist and chemist.

4.2.1 Assay design:

1. **Assay technology:**
    a) Use red-shifting fluorophores over others
    b) Use dual-luciferase-based assays over one luciferase assays
    c) Use TR-FRET over FRET

2. **Assay readout:**
    a) Apply bandpass filters to exclude auto-fluorescence
    b) Use functional readouts that are more resistant to simple interferences
    c) Include cell counts in imaging assays to reflect treatment toxicity

3. **Assay buffer:**

a) Include nonionic detergents and/or a decoy protein to prevent colloidal aggregation
b) Include scavenging reagents to prevent non-specific chemical reactivity

4. **Quality of test compounds**:
    a) Ensure purity and stability of the control compounds
    b) Use high-quality chemical libraries of compound purity > 85%

5. **Quality of reagents**:
    a) Ensure batch-to-batch consistency for assay reagents, especially for genetically modified cells and purified proteins
    b) If not obtained directly from the supplier, the quality and integrity of cell lines should be proven through short tandem repeat (STR) profiling[6]
    c) Ensure absence of mycoplasma infection
    d) Verify correct sequence of plasmids used for transient transfections

6. **Quality of assays**:
    a) Establish assay SOP and evaluate assay quality for every batch of experiment. The minimum Z´ factor[7] cut-off ($Z' > 0.5$) is applied per default for reader-based biochemical and cellular screens. A $0.4 < Z' < 0.5$ can be accepted for cellular assays.
    b) Use a positive (maximum effect) control and negative (vehicle) control for normalization within each test plate.
    c) To track assay performance over time include the same QC compound in each batch of experiments, ideally as a full concentration response curve or as one concentration at IC_{50}/EC_{50} and plot performance over time.

4.2.2 Hit Compound evaluation:

1. **Quality of hit compounds**:
    a) Perform hit compound identity and purity analysis with standard methods (HPLC_MS (M+1) at the Compound Center. Ideally a compound should have purity > 85% but less can be accepted, Compounds that fails to be analyzed with these methods must be dealt with individually, eg no chromophore in HPLC analysis or no ionization in MS.
    b) Obtain fresh material for retest. Re-synthesize, re-dissolve from solid, purchase, or by any other means get fresh material for retest.

2. **Test compounds in counter screens**
    a) Run counter assays to remove readout specific artefacts such as color, auto florescence, coupled enzyme inhibition or other relevant aspects.
    b) Run phospholipidosis assay
    c) Run cell viability (Cell Titer Glo, imaging based cell count)

d) Run redox assays; in-house modification of a litterature procedures Rezazurin[9], and/or HRP/Phenol red assay[10], add or remove reducing agent depending on previous assay setup.
e) Run aggregation assay

3. Assess direct binding of compounds to target proteins.
a) Run target-binding assay (SYPRO Orange Thermal Shift Assay, Nano-DSF)
b) Run target engagement assay (CETSA)[11]

4. In silico hit evaluation
a) Run structural filters for PAINS, REOS, and aggregation
b) Compound clustering and visual inspection of structures by the chemistry team
c) Perform hit compound literature searches in Cortellis, SciFinder, and possibly any other drug-target databases of choice
d) Analyze any preliminary SAR from the screening set
e) Select analogs from in-house libraries, commercial sources, or possibly to synthesize a few for further SAR studies. In general, hits that are part of a chemical hit series are preferred over singletons
f) Perform cheminformatics analysis of activities identified from historical in-house results

5. Orthogonal and secondary assays
a) Test hits in orthogonal assays available in the infrastructure user's (principle investigator') lab
b) Test hits in secondary assays available in the infrastructure user's (principle investigator') lab
c) Adressing target selectivity and ways to assess this is always of outmost importance in every project discussion.

5 Further applicable documents

Literature references

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